



A difference attention ResNet-LSTM network for epileptic seizure detection using EEG signal

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ABSTRACT

Epileptic seizures can affect the patient's physical function and cause irreversible damage to their brain. It is vital to detect epilepsy seizures in time and give patients antiepileptic medical treatment. Hybrid deep learning models, which combine convolutional neural network and recurrent neural network, have better epileptic seizure detection performance as they could simultaneously extract spatial and temporal features. However, the existing hybrid deep learning models still have the following two weaknesses. Firstly, they directly input the raw electroencephalogram signals, where the epilepsy seizure information is limited. Secondly, some characteristic information is extracted in the feature map, distracting the attention of deep learning model. To address these issues, this paper proposes a difference attention ResNet-LSTM network (DARLNet). The proposed model uses a residual neural network (ResNet) and a long short-term memory network (LSTM) to capture spatial correlations and temporal dependencies, respectively. Besides, a difference layer is developed to automatically mine additional epileptic seizure information. Moreover, the channel attention module is introduced to make the model focus on seizure-relevant information. Several groups of experiments are conducted to evaluate the performance of DARLNet based on the Bonn Electroencephalogram dataset, which verifies the superiority of DARLNet on the two-category and five-category epileptic seizure detection tasks.

1. Introduction

Epilepsy is a common chronic neurological disease caused by the abnormal discharge response of brain neurons [1,2]. Usually, patients with epilepsy will lose consciousness due to transient brain dysfunction, local or general convulsions and fainting symptoms during the onset of illness [3,4]. There are approximately 65 million patients with epilepsy in the world, with about 2 million new cases each year [5]. In addition, frequent disease will cause irreversible damage to the patient's brain, severely affect the patient's physical function, and cause psychological damage [6]. Therefore, it is greatly significant to diagnose epilepsy patients in time and give antiepileptic medical treatment.

In general, neurologists detect epilepsy and locate the diseased area by analyzing the shape and amplitude of abnormal bands of brain waves. However, relying on neurologists to diagnose epilepsy through visual observation has the following shortcomings. Firstly, visual observations rely on expert experience and professional knowledge. Secondly, the acquisition frequency of electroencephalogram (EEG) signal is high and the collected data volume is large, which requires a long time for visual observation [7]. Besides, long-term visual observation will affect the accuracy of neurologists' detection, resulting in serious consequences such as misdiagnosis and missed diagnosis. In addition,

EEG signals are easily affected by environmental noise, which increases the difficulty of visual observation detection. Therefore, it is necessary to develop a computer-assisted system to help neurologists detect epilepsy.

Machine learning (ML) used to be the most commonly used epileptic seizure detection method. However, ML methods require the manual design of feature extractors to capture epileptic seizure features, which is hard to extract discriminative features [8,9]. In recent years, the rapid progress of neural network theory and parallel computing technology has made significant breakthroughs in deep learning (DL). DL technologies have been successfully applied in image recognition [10], speech recognition [11], and emotion recognition [12]. Furthermore, its applications in the medical field have also emerged [13]. Deep learning techniques are developed from machine learning methods and usually have a multi-layer structure, and the use of activation functions enables them to have stronger nonlinear characterization capabilities [14–16]. DL technologies can automatically capture effective features from EEG signals and achieve end-to-end epileptic seizure detection [17,18]. The application of DL technologies in epileptic seizure detection could reduce the workload of neurologists, promote work efficiency and improve detection accuracy. Therefore, this

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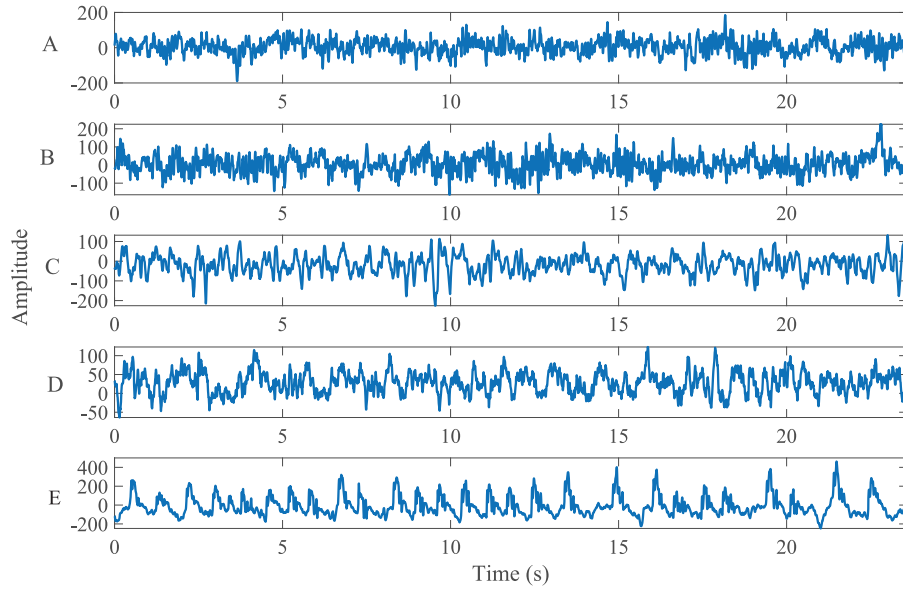


Fig. 1. Example of EEG signals from each of the five sets (A, B, C, D, and E). From top to bottom: EEG signals A to EEG signals E.

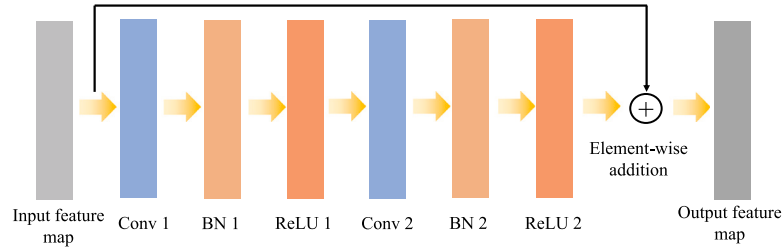


Fig. 2. The structure of residual block.

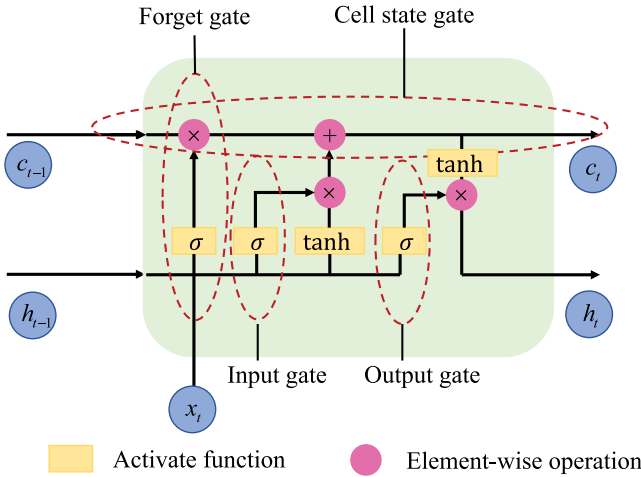


Fig. 3. The structure of LSTM. The basic neurons of LSTM include forget gate, input gate, cell state gate and output gate.

paper is interested in investigating the DL-based epileptic seizure detection method. The DL-based epileptic seizure detection models can be roughly divided into three categories: convolutional neural network (CNN)-based models, recurrent neural network (RNN)-based models, and hybrid models [19–22].

CNN-based models extract features from EEG signals through convolution and pooling operations. Several kinds of CNN networks, including multi-layer CNN, one-dimensional convolutional neural network

(1DCNN), three-dimensional convolutional neural network (3DCNN), have been adopted for epileptic seizure detection. Acharya et al. [23] proposed a 13-layer CNN to distinguish normal EEG signals, epileptic intermittent signals, and epileptic onset signals. Antoniadis et al. [24] designed a four-layer CNN to detect epileptiform discharges in intracranial EEG. Elakkiya et al. [25] developed an epileptic seizure detection model based on a 1DCNN model. Wei et al. [26] introduced a 3DCNN for epileptic seizure detection using multi-channel EEG signals. Shoji et al. [27] proposed a multi-channel CNN network to detect abnormal patterns of EEG signals.

RNN-based models are often used to process sequential data and are widely applied in natural language processing and machine translation. Long short-term memory network (LSTM) is a variant of RNN. By introducing the gated mechanism, LSTM effectively addresses the gradient vanishing problem of the traditional RNN model [28,29], and is suitable for long-term dependency handling. Abbasi et al. [30] established a two-layer LSTM for epileptic seizure detection. Manohar et al. [31] used short-time Fourier transform to preprocess the EEG signals and then used the bidirectional LSTM to conduct epilepsy seizure detection.

Recently, researchers have attempted to take advantage of both CNN and RNN models [32]. Many hybrid models have been developed. Xu et al. [33] proposed a 1DCNN-LSTM model to identify epileptic seizures, which achieved a 5% improvement on F1-score over a CNN model. Liu et al. [34] proposed an epilepsy detection method combined CNN and Bi-LSTM, and the proposed approach performed relatively better compared with four variants of CNN and a LSTM model.

Hybrid model-based epileptic seizure detection has been the recent research trend. As can be seen in the literature review, hybrid models generally perform better because they can extract spatial and temporal epilepsy seizure information simultaneously. Therefore, this

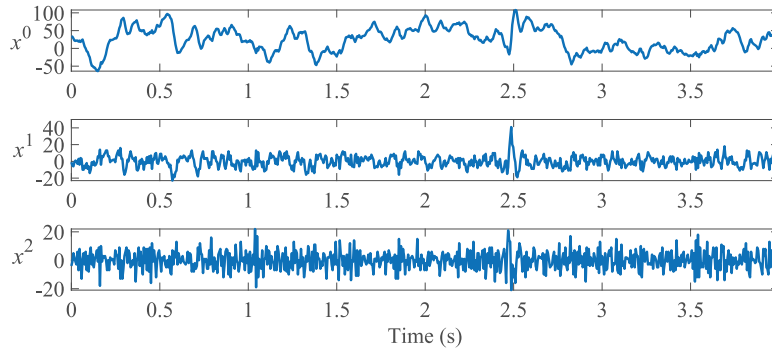


Fig. 4. The raw EEG signal and the corresponding first-order and second-order backward difference signals, where x^0 is the original EEG signal, x^1 is the first-order backward difference signal, and x^2 is the second-order backward difference signal.

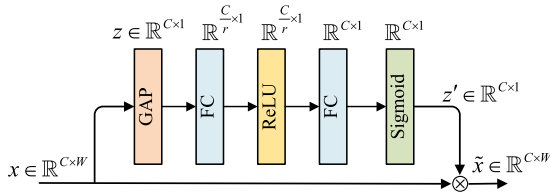


Fig. 5. The structure of channel attention module.

paper also constructs a hybrid model for epilepsy seizure detection. In addition, the above-mentioned hybrid methods still have the following two weaknesses. On the one hand, most studies directly use raw EEG signals as the input, and the implied epilepsy seizure information is limited. On the other hand, some irrelevant and redundant information in the feature map will distract the network's attention, which negatively affects epileptic seizure detection. This paper proposed a difference attention ResNet-LSTM network (DARLNet) to address the above problems. The proposed model adopts ResNet and LSTM to capture spatial correlations and temporal dependencies, respectively. In addition, a difference layer, which calculates the first-order and second-order backward difference signals, is developed to provide extra epileptic seizure information. This mechanism mitigates the first weakness to some extent. Moreover, the channel attention module is introduced to highlight the seizure-relevant information to address the second weakness. Based on the Bonn EEG dataset, two-category and five-category detection tasks are carried out to verify the proposed method's effectiveness. The results show that the proposed DARLNet outperforms several epilepsy seizure detection methods.

The rest of this paper is organized as follows: The used dataset, the structure of ResNet and LSTM, and the proposed methods are described in Section 2. Section 3 presents the experimental results and result analysis. Finally, conclusions are given in Section 4.

2. Materials and methods

2.1. Dataset

2.1.1. The University of Bonn dataset

This paper utilizes the EEG dataset of the Epilepsy Center at the University Hospital of Bonn to conduct classification experiments. As the first publicly accessible database, the Bonn EEG dataset is the most widely used data in the field of epileptic seizure detection [35]. This dataset contains five subsets (denoted A-E), and each of the subsets includes 100 single-channel EEG fragments of 23.6-s duration. Data are recorded using a standard 10–20 electrode placement system, and all segments in the dataset are acquired at a frequency of 173.61 Hz, so each fragment contains 4097 data points. More details on the subsets of the Bonn EEG dataset are as follows:

Subsets A and B: Both are collected from healthy people. EEG recordings are collected from the scalp. A is the EEG recordings collected when the volunteers opened their eyes, while B is the EEG recordings collected when the volunteers closed their eyes. Subsets C, D, and E: These EEG recordings are collected from five epilepsy patients. Among them, subsets C and D are collected from the epileptic interval, the collection location of subset C is the opposite side of the lesion, and that of subset D is inside the lesion. Subset E is collected from the epileptic period, and the collection location is inside the lesion.

Fig. 1 shows a sample of randomly selected EEG recordings from subsets A, B, C, D, and E. It can be seen that the amplitude of the EEG signal is stronger than that of other periods when the epileptic seizure occurs. We divide each EEG recording into 23 signal segments, and each segment contains 178 data points. Finally, for 500 EEG recordings, 11,500 signal segments are generated. We randomly select 90% of signal segments as the training set, and the other signal segments are used as the testing set.

2.1.2. The Hauz Khas Delhi dataset

This dataset is collected from Neurology & Sleep Centre [36]. The sampling frequency of the collected data is 200 Hz. During the acquisition process, gold plated scalp EEG electrodes were placed according to 10–20 electrode placement system. Filtering was performed between 0.5 and 70 Hz, and the collected signals were then divided into three categories: preictal, interictal, and ictal. The length of each EEG time series data sample is 1024.

2.2. The structure of ResNet and LSTM

2.2.1. ResNet

ResNet is an emerging DL approach that has attracted much attention in recent years [37,38]. The residual block is the basic component of ResNet. As shown in Fig. 2, the residual block consists of two convolutional (Conv) layers, two batch normalization (BN) layers, two rectifier linear units (ReLU) and an identity shortcut. The main difference between ResNet and CNN-based methods is that the introduction of identity shortcut enables more gradient information to flow through, which facilitates the update of parameters [39].

2.2.2. LSTM

LSTM is a well-known variant of RNN, which alleviates the gradient vanishing problem of RNN [40,41]. Fig. 3 presents the structure of an LSTM unit. Each LSTM unit is composed of four gates: forget gate, input gate, cell state gate, and output gate. The forget gate controls the information from the previous moment to be kept, input gate controls the current input information to be saved, cell state gate controls information to be remembered in the cell, and output gate controls the learned information to generate the output of the LSTM unit. Through using the gated mechanism, LSTM can effectively capture the long-term temporal information [42].

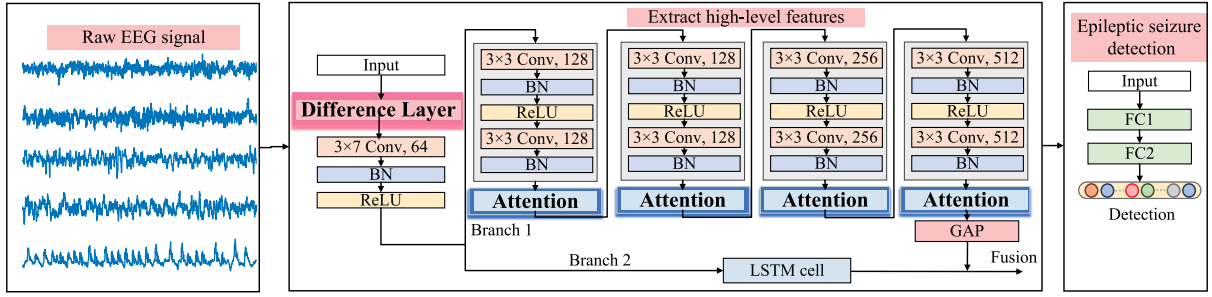


Fig. 6. The framework of the proposed DARNET model. First, the raw EEG signals are directly fed into the DARNET model, and the difference layer calculates the first-order and second-order backward difference signals to obtain additional seizure information. Then, the ResNet and LSTM extract the localized correlation features and temporal features, respectively. Moreover, the introduced channel attention module highlights the epileptic seizure characteristic adaptively. Finally, the fully connected layer fuses the learned high-level epileptic seizure features and gives the final epileptic seizure detection result.

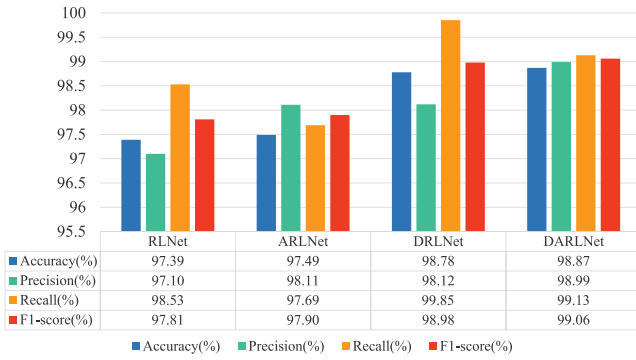


Fig. 7. Two-category detection task ablation experiment results.

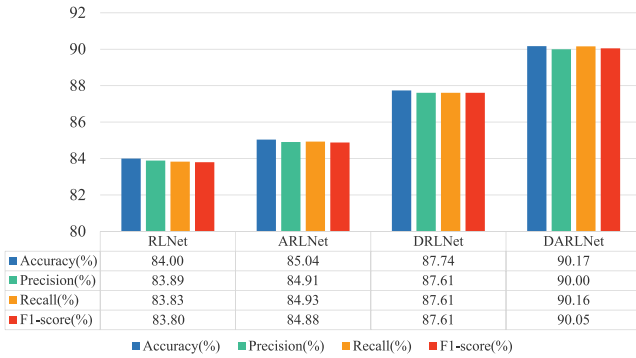


Fig. 8. Five-category detection task ablation experiment results.

2.3. The proposed methods

2.3.1. The difference layer

Most previous studies directly input the raw EEG signals into the DL model. However, the epileptic seizure features in the raw EEG signals are limited and hard to capture, resulting in poor robustness of the epileptic seizure detection model. Under this background, we aim to provide supplementary information for the DL model and improve the accuracy of epileptic seizure detection. Motivated by this, we developed a difference module for feature enhancement by calculating the first-order and second-order backward difference signals of the raw EEG signals, and then the generated backward difference signals together with the raw EEG signals are taken as the input of the DL model.

Fig. 4 shows a raw EEG signal and the obtained backward difference signals from the proposed difference module. It can be seen that the first-order and second-order backward difference signals can reflect the change signal features and rate of signal change features, respectively.

For the input raw EEG signal ($x = [x_1, x_2, \dots, x_N]$), the computation process of the difference layer can be expressed as:

$$x = x^0 = \{x_1^0, x_2^0, \dots, x_N^0\} \quad (1)$$

$$x^1 = \{x_1^1, x_2^1, \dots, x_N^1\}, x_j^1 = x_j^0 - x_{j-1}^0, x_0^0 = 0, j = 1, 2, \dots, N \quad (2)$$

$$x^2 = \{x_1^2, x_2^2, \dots, x_N^2\}, x_j^2 = x_j^1 - x_{j-1}^1, x_0^1 = 0, j = 1, 2, \dots, N \quad (3)$$

$$y = \parallel_{k=0}^2 x^k \quad (4)$$

where x^1 and x^2 indicate the first-order and second-order backward difference signal, $\parallel$ denotes the concatenate operation, y represents the output of the difference layer [43]. Note that the obtained backward difference signals are taken as channels of the input data, respectively, so the dimension of the output y is $3 \times N$.

2.3.2. Channel attention module

When using the convolutional-based method to extract features from the input signals, some irrelevant and redundant information are inevitably extracted in the feature map, which has a negative effect on epileptic seizure detection. This paper introduces the channel attention module into the model to adaptively enhance the effect of useful information and suppress that of irrelevant and redundant information [44].

Fig. 5 illustrates the structure of the channel attention module. This module can adaptively adjust the importance of feature channels by assigning different weights to each channel of the feature map. Suppose that $x \in \mathbb{R}^{C \times W}$ is the input of the channel attention module, where C and W denote the number of input channels and the length of input, respectively, x_i denotes the i -th channel of x . The calculation process of the channel attention module can be divided into two steps.

Firstly, the inputted feature map is compressed into channel-wise statistics vector z by a global average pooling layer (GAP) along the temporal dimension, and the calculation of the i -th element z_i can be defined as (Note that z_i is a scalar):

$$z_i = \text{Avgpool}(x_i) = \frac{1}{1 \times W} \sum_{j=1}^W x_i(j) \quad (5)$$

Secondly, computing the channel attention vector z' using the fully connected layer (FC), which can be expressed as:

$$z' = \sigma(W_2 \delta(W_1 z)) \quad (6)$$

where δ and σ denote the ReLU function and Sigmoid function, respectively. $W_1 \in \mathbb{R}^{C \times \frac{C}{r}}$ and $W_2 \in \mathbb{R}^{\frac{C}{r} \times C}$ are learnable weights implemented by the fully connected layers, where r indicates the dimensionality reduction ratio. The i -th value z'_i of channel attention vector represents the i -th channel importance of feature map x , and the channel attention vector adjusts the feature importance using the following equation:

$$\tilde{x} = x \otimes z' = [x_1 z'_1, x_2 z'_2, \dots, x_c z'_c] \quad (7)$$

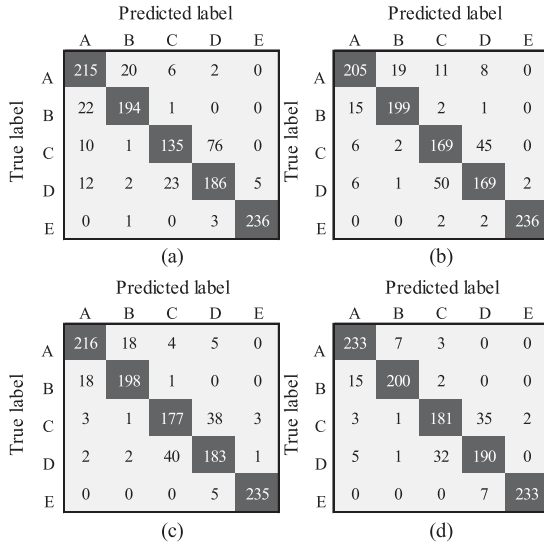


Fig. 9. Confusion matrix of models (a) RLNet, (b) ARLNet, (c) DRLNet, and (d) DARNet. The confusion matrix is a specific matrix used to visualize the performance of an algorithm, with each column representing the predicted value and each row representing the actual category.

Finally, in the generated feature map $\bar{x}$, larger vectors are assigned to channels with more seizure-relevant information, and smaller vectors are assigned to channels with more redundant information. This process is done automatically in network training. Therefore, the channel attention module can effectively highlight useful information, which is beneficial for epileptic seizure detection.

2.3.3. Difference Attention ResNet-LSTM network (DARNet)

Based on the above two implements, a Difference Attention ResNet-LSTM network (DARNet) is formed, as shown in Fig. 6. Firstly, the DARNet model takes the raw EEG signal as the input, and the developed difference layer calculates the backward difference signals to provide helpful extra information for the model. Then, the obtained backward difference signals along with the raw EEG signal are fed into the first residual blocks of ResNet18 to extract the initial features. Moreover, the learned initial features are inputted into two parallel branches. The first branch consists of four residual blocks, which use learnable kernels to capture local correlation characteristics from the input. The channel attention module is introduced and connected to the residual block to enhance feature discriminability. The second branch adopts the LSTM cells to model the temporal dependencies. Finally, the extracted high-level epileptic seizure features are concatenated and fed into the FC layers to perform feature fusion and epileptic seizure detection. The detailed structure of DARNet and its output feature size of each layer are summarized in Table 1. The first, second and third parameters in Conv(.) and Residual block (.) denote the number of convolution kernels, the size of convolution kernels, and the stride size, respectively. The parameter in Maxpool(.) indicates the kernel size of max-pooling. In addition, it should be mentioned that the zero padding technique is adopted in all the convolutional layers.

3. Experiment and results

In this section, experiments are conducted to evaluate the effectiveness of the proposed DARNet model based on the Bonn EEG dataset. The detailed descriptions of the experimental settings and evaluation metrics are given below.

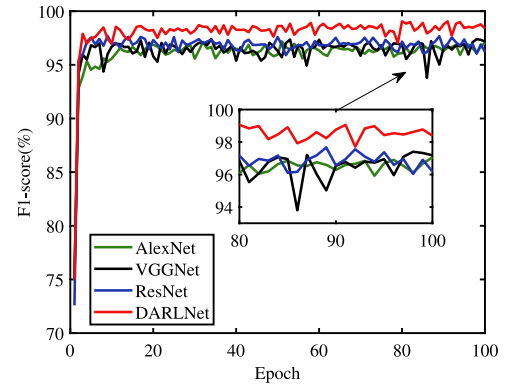


Fig. 10. Testing F1-score of the four models on the two-category detection task.

Table 1

Detailed structure of the proposed DARNet.

Numbers	Type of layer	DARNet	Output feature size
1	Backbone 1	Input	$1 \times 178 \times 1$
2	Backbone 2	The Difference layer	$3 \times 178 \times 1$
3	Backbone 3	Conv(64,7,/2)	$64 \times 89 \times 1$
4	Backbone 4	BN,ReLU,Maxpool(2)	$64 \times 45 \times 1$
5	Branch 1-1	Residual block(64,3,/1)	$64 \times 45 \times 1$
6	Branch 1-2	Residual block(128,3,/2)	$128 \times 23 \times 1$
7	Branch 1-3	Residual block(256,3,/2)	$256 \times 12 \times 1$
8	Branch 1-4	Residual block(512,3,/1)	$512 \times 12 \times 1$
9	Branch 1-5	GAP	512
10	Branch 2	LSTM cell	64
11	Backbone 5	FC1	256
12	Backbone 6	FC2	2 or 5

3.1. Experimental settings

This article studies two kinds of epileptic seizure detection tasks: two-category detection and five-category detection. To be specific, the two-category detection task classifies the seizure condition and the normal condition. In contrast, the five-category detection task refers to distinguishing the seizure condition and the four non-onset conditions.

To verify the effectiveness of the proposed DARNet, ablation experiments and comparative experiments are carried out. All DL models are implemented based on the Pytorch architecture. A workstation with the Win 10 operating system, Intel Xeon Gold 6226 CPU, and Tesla P100 GPU is used for model training and testing. The learning rate and batch size are set to 0.001 and 50, respectively. The batch normalization layer is added behind each convolutional layer. The dropout rate is set to 0, as no overfitting phenomenon was observed in the experiments. The cross-entropy function is adopted as the loss function, and the adam optimizer is taken as the optimizer. All DL models are finally trained for 100 Epochs, and the best result of all training rounds is taken as the final result.

3.2. Evaluation metric

In this study, we use four typical classification indicators in order to evaluate the epileptic seizure detection performance of the proposed model. The four typical classification indicators are accuracy, precision, recall, and F1-score. Among them, F1-score is a comprehensive metric that measures the classification performance. These indicators are defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

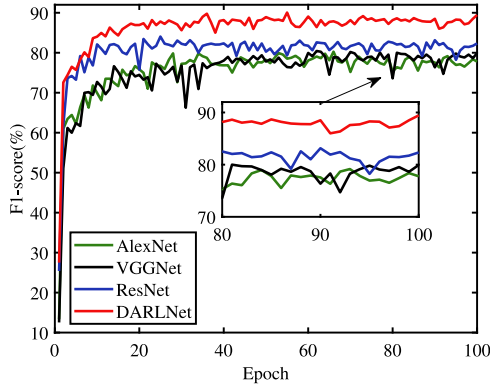


Fig. 11. Testing F1-score of the four models on the five-category detection task.

$$F1\text{-score} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (11)$$

where TP and TN represent the number of true positive samples and true negative samples, respectively. And FP and FN denote the number of false positive samples and false negative samples, respectively. A larger value of accuracy, precision, recall, and F1-score mean better epileptic seizure detection performance.

3.3. Experimental results

3.3.1. Ablation studies

This section carries out ablation studies to verify the effectiveness of the developed difference layer and the introduced channel attention module. Three degraded versions of DARNet models are established, which are RLNet, ARLNet and DRLNet models. Among them, the ARLNet model does not have the difference layer, the DRLNet model does not have the channel attention module, and the RLNet model gets rid of both. Experimental results are shown in Figs. 7 and 8.

We can see that the performance of the DARNet model is better than that of other compared models as its F1-score reached 99.06% from Fig. 7. Furthermore, from the comparison between RLNet and DRLNet as well as the comparison between ARLNet and DARNet, it can be seen that the proposed difference layer can significantly enhance the model performance. This is because the difference layer provides extra helpful information for the model to capture more features associated with epileptic seizures. In addition, from the comparison between RLNet and ARLNet along with the comparison between DRLNet and DARNet, we find that the introduced channel attention module can slightly improve the F1-score of the model. A possible reason is that the F1-scores of RLNet and DRLNet are already high, and the effect of highlighting useful information is not obvious.

From Fig. 8, it can be seen that the F1-score of DRLNet is 3.81% higher than that of RLNet, while the F1-score of DARNet is increased by 5.17% compared with ARLNet.

The reason seems straightforward: the proposed difference layer calculates the backward difference signals for feature enhancement of the DRLNet and DARNet model. In addition, from the comparison between RLNet and ARLNet and the comparison between DRLNet and DARNet, we observe that the introduced channel attention module is also helpful for epileptic seizure detection. The F1-score of ARLNet and DARNet is about 1% higher than that of RLNet and DRLNet, respectively. This is because the integrated channel attention module enables the model to focus on essential epilepsy characteristics adaptively, thus the model can encode high-level features and achieve more accurate classification results. The above observations confirm that the proposed difference layer and the introduced channel attention module are effective for epileptic seizure detection.

Table 2

The classification results of EEG signals in the Hauz Khas Delhi dataset.

	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)
preictal-ictal	100.00	100.00	100.00	100.00
interictal-ictal	100.00	100.00	100.00	100.00

Table 3

The performance of Alexnet, Vgg16, Resnet, 1DCNN-LSTM and the proposed DARNet model on the two-class classification task.

Models	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)
AlexNet	96.17	97.39	96.28	96.83
Vgg16	96.61	97.68	96.70	97.19
ResNet	97.04	97.99	97.67	97.53
1DCNN-LSTM	99.39	98.39	98.79	98.59
DARNet	98.87	98.99	99.13	99.06

Table 4

The performance of Alexnet, Vgg16, Resnet, 1DCNN-LSTM and the proposed DARNet model on the five-class classification task.

Models	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)
AlexNet	79.48	79.37	79.24	79.12
Vgg16	80.52	80.41	81.15	80.03
ResNet	84.17	84.08	84.07	84.04
1DCNN-LSTM	82.00	81.78	81.70	81.56
DARNet	90.17	90.00	90.16	90.05

To analyze the epileptic seizure detection performance of each condition, the confusion matrix of models RLNet, ARLNet, DRLNet, and DARNet are shown in Fig. 9. The row and column in these confusion matrices indicate the predicted label and true label. It can be seen that the main error comes from misclassifying samples from subsets C and D, and the wrong classification condition of DARNet is relatively better than that of other models.

3.3.2. Classification results of the Hauz Khas Delhi dataset

In order to verify the validity of the proposed model, we perform the validation on the Hauz Khas Delhi dataset. Two cases are considered in this work: preictal-ictal and interictal-ictal. Table 2 lists the classification results of EEG signals in the Hauz Khas Delhi dataset using the proposed model. The ictal and preictal EEG signal segments are classified with 100.00% accuracy, precision, recall, and F1-score. In addition, the ictal and interictal EEG signal segments are also classified with 100.00% accuracy, precision, recall, and F1-score. The proposed method shows good performance in detecting ictal and interictal segments.

3.3.3. Comparative experiment

To further evaluate the validity of the proposed method, this subsection compared DARNet with four epileptic seizure detection methods, which are ResNet, Vgg16, AlexNet and 1DCNN-LSTM. As aforementioned, two kinds of experiments are conducted, i.e., two-category detection and five-category detection tasks. Experimental results are recorded in Tables 3 and 4.

From Table 3, we find that DARNet outperforms other compared models, as the F1-score of DARNet is 1.53%, 1.87%, 2.23% and 0.47% higher than that of ResNet, Vgg16, AlexNet, and 1DCNN-LSTM. Fig. 10 shows the F1-score for four models (AlexNet, VGGNet, ResNet, and DARNet) on the two-category detection task. Obviously, the proposed DARNet model is superior to the other compared model. Besides, the performance of ResNet is better than that of Vgg16 and AlexNet. It may be ascribed that the used residual architecture could make more useful epileptic features flow through. 1DCNN-LSTM also obtains outstanding results, and the similarity between 1DCNN-LSTM and DARNet is that they both extract temporal features and correlation features from the input.

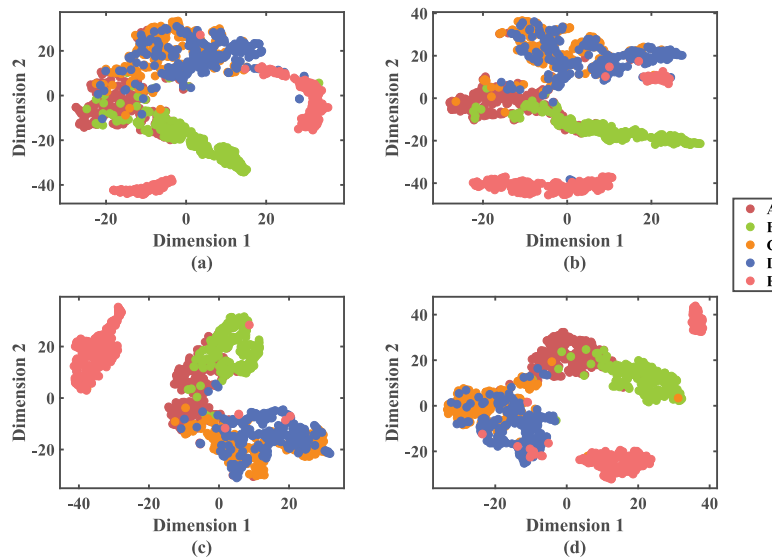


Fig. 12. 2-D visualizations of features at the final layer in (a) AlexNet, (b) VGGNet, (c) ResNet, and (d) DARLNet. Among them, dark red represents dataset A, green represents dataset B, orange represents dataset C, blue represents dataset D, and bright red represents dataset E. The less overlap between points of different colors, the better the classification effect of the model.

Table 5

The performance of the proposed method and several existing methods. (A: Collected from volunteers with their eyes open. B: Collected from volunteers with their eyes closed. C: Collected from patients during the epileptic interval, and the collection location is opposite side the lesion. D: Collected from patients during the epileptic interval, and the collection location is inside the lesion. E: Collected from patients during the epileptic period, and the collection location is inside the lesion.).

Classification task	Authors	Years	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)
A-E	The proposed method		100.00	100.00	100.00	100.00
	Tajmirriahi et al. [45]	2021	99.00	\	99.00	\
	Ramos-Aguilar et al. [46]	2020	100.00	\	\	\
	Siuly et al. [47]	2019	99.50	\	100.00	\
	The proposed method		100.00	100.00	100.00	100.00
B-E	Tajmirriahi et al. [45]	2021	99.50	\	100.00	\
	Ramos-Aguilar et al. [46]	2020	99.80	\	\	\
	Siuly et al. [47]	2019	99.00	\	99.09	\
	Deng et al. [48]	2018	97.01	\	\	\
	The proposed method		99.78	100.00	99.56	99.78
C-E	Tajmirriahi et al. [45]	2021	99.60	\	98.00	\
	Ramos-Aguilar et al. [46]	2020	99.25	\	\	\
	Siuly et al. [47]	2019	98.50	\	98.18	\
	The proposed method		99.57	99.12	100.00	99.56
D-E	Tajmirriahi et al. [45]	2021	99.20	\	98.80	\
	Ramos-Aguilar et al. [46]	2020	97.75	\	\	\
	Siuly et al. [47]	2019	97.50	\	98.09	\
	The proposed method		98.17	97.46	97.50	97.46
AB-CD-E	Acharya et al. [23]	2018	88.67	\	95.00	\
	The proposed method		90.17	90.00	90.16	90.05
A-B-C-D-E	Xu et al. [32]	2020	82.00	81.78	81.70	81.56

According to Table 4, we observe that the accuracy, precision, recall, and the F1-score of DARLNet are 90.17%, 90.00%, 90.16%, and 90.05%, respectively. Obviously, it is better than those of ResNet, Vgg16, AlexNet, and 1DCNN-LSTM. To see the superiority of the proposed DARLNet model, the F1-score of four tested models on the five-category detection task is displayed in Fig. 11. It can be observed that DARLNet obtains the highest F1-score throughout the whole training.

The t-distributed stochastic neighbor embedding technology (t-SNE) [39], a well-known unsupervised dimension reduction approach, is adopted to understand whether the extracted features are discriminative or not. Fig. 12 shows the t-SNE visualizations of the last layer output of models ResNet, Vgg16, AlexNet, and 1DCNN-LSTM in 2-D space. As shown in Fig. 12(a), (b), and (c), the generated 2-D observations under different health conditions suffer serious overlapped, especially for the datasets A, C, and D. The 2-D features of dataset E are distributed in two regions, due to the EEG signal are collected from

different patients and have different seizure characteristics. According to Fig. 12(d), it can be seen that the 2-D features of the same condition are more concentrated, and fewer 2-D features are overlapped. The above experimental results show that the proposed difference layer and the introduced attention mechanism are beneficial for epileptic seizure recognition, and the DARLNet could achieve excellent recognition performance.

Finally, Table 5 recorded the performance of the proposed DARLNet and some of the recently proposed method [23,32,45–48] on the Bonn dataset. It can be seen that our proposal achieves the state-of-the-art epilepsy detection results in binary classification tasks (A-E, B-E, C-E, D-E) and multi-classification tasks (AB-CD-E, A-B-C-D-E). Especially for multi-classification tasks, the detection results of the proposed method is significantly improved compared to that of the comparison method, which proves the superiority of DARLNet.

4. Conclusion

This paper proposes a DARNet for epileptic seizure detection, which combines the advantages of CNN-based network and RNN-based network by using Resnet and LSTM to extract spatial correlations and temporal dependencies in parallel. In addition, a difference layer is proposed to provide extra useful information by calculating the first-order and second-order difference signals. Besides, the channel attention module is introduced to adaptively adjust the importance of feature channels and highlight seizure-relevant information. Based on the Bonn EEG dataset, the effectiveness of the proposed DARNet for epileptic seizure detection is verified by the two-category and five-category detection tasks. Ablation experiment results reveal that the proposed difference layer and the introduced channel attention module can effectively enhance the model performance. Comparative experiments between DARNet and several epileptic seizure detection methods were conducted, and DARNet obtained the highest F1-score of 99.06% and 90.05% on the two tasks, respectively. Moreover, the proposed method can also be applied to the detection of other similar diseases, which has important significance for clinical detection and analysis.

The limitation of this paper is that experiments were mainly performed on the single-channel dataset, and we will further study epileptic seizure detection using multichannel EEG recordings in the future.

Declaration of competing interest

No author associated with this paper has disclosed any potential or pertinent conflicts which may be perceived to have impending conflict with this work. For full disclosure statements refer to <https://doi.org/10.1016/j.bspc.2023.104652>.

Data availability

Data will be made available on request.

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